



Machine Learning

Quiz - 1 (20 Marks) (Duration 60 min)



Beginners (8 marks)

1. Shyam read about Generalized linear model (GLM) and made some assumptions, help Shyam by commenting True/False on each statement. Justify your answers.
- A. Target variable are independently distributed. B. Target Variable must normally distributed C. Non linear relationship between link function and explanatory variables D. Errors must be normally distributed

True: This is a fundamental assumption for GLMs. This allows the total likelihood of the data to be calculated as the product of the individual likelihoods for each observation.

False: This is the primary assumption that GLMs are designed to relax. Target variable's distribution can be any member of the exponential family.

False: GLM explicitly assumes a linear relationship between the output of the link function and the explanatory variables.

False: Assumption not valid for all GLMs.

2. Probit regression is an alternative to logistic regression that assumes an underlying...

A. Poisson B. Gamma C. Normal D. Binomial

C

3. You flip a coin 10 times and get 7 heads. What is the MLE for the probability of getting heads?

A. 3 B. 1.4 C. 7 D. 0.7

D

4. If logistic regression outputs probabilities bounded between 0 and 1, why can't we simply use linear regression and clip the outputs to range? Clipping creates discontinuous gradients at the boundaries (0 and 1), making optimization difficult. Linear regression assumes Gaussian noise which is inappropriate for binary outcomes. The sigmoid function provides smooth, differentiable transitions and has theoretical justification through maximum likelihood estimation for Bernoulli distributions

5. If $h_b(x_i) = 1$ when $x_i b > 0$, what class would be predicted when $x_i b = 0$ exactly? When $x_i b = 0$ exactly, the point lies on the decision boundary. The hypothesis function would predict Class 0 (since it's "0 otherwise"), but this is an edge case that rarely occurs in practice with real-valued data. The sigmoid probability would be exactly 0.5, indicating maximum uncertainty.

6. How can we reduce/avoid overfitting from a function approximation perspective? Reducing the space; Making the choice of f^* less dependent on data

7. As the amount of training data becomes very large, the difference between the MAP and MLE estimates typically... (and Why?)

A. Increases B. Stays the same C. Decreases D. Can't be determined C; With enough data, the likelihood term overwhelms the prior, and the MAP estimate converges to the MLE estimate.

8. Explain Berkson's & Simpson's Paradox.

Berkson's paradox is a result in statistics, very closely related to Simpson's paradox, that demonstrates that two values can statistically be negatively correlated even when they appear positively correlated in the population. Example: In our class, being taller may appear to be positively correlated with being good at math. However, statistically, a student's height and math skills are not correlated, and being taller doesn't make someone better or worse at math. It might be simply the case that in our dataset taller students tend to be older and have studied more math. Hence, Berkson's paradox is a particular kind of selection bias, caused by systematically observing some events more than others. Berkson's paradox, and selection bias in general, appear in almost every field of research, particularly epidemiology and economics.

Do you know that Simpson's Paradox is a fascinating statistical phenomenon where the relationship between two variables can completely change when a third variable is considered, leading to contradictory conclusions. For example, in a data set, you might see a positive trend between variables X and Y. However, when you split the data into subgroups based on a third variable, Z, the trend might reverse or disappear. Key takeaways: Always consider potential lurking variables that could influence your results. Be cautious with aggregated data; sometimes the real insights lie in the subgroups. Use visualizations to better understand the relationships within your data. The Wikipedia article on Simpson's Paradox offers a great visualization, emphasizing the importance of thorough analysis to avoid such pitfalls.

Intermediate (2 marks)

1. A programmer writes a set of explicit, hard-coded rules to determine if an email is spam (e.g., "IF the email contains the words 'free money', THEN mark as spam"). A machine learning engineer, however, feeds a model 100,000 emails that are already labeled as "spam" or "not spam" and lets the model learn the patterns itself. Why is the machine learning engineer's approach considered inductive learning, while the programmer's approach is not?

The ML engineer's approach is inductive because the model is deriving a general rule (what makes an email spam) from specific examples (the labeled emails). It induces the pattern from the data. The programmer's approach is based on pre-defined rules and is not learning from data; it's simply executing commands.

2. With learning rate $\eta = 0.003$ and $W_{old} = 0.897$ derive and write new updated weight W_{new} using this loss function equation $L(x, \theta) = \left(\frac{1}{Y_{actual} - Y_{predicted}} \right)$.

Not possible, because when $Y_{actual} == Y_{predicted}$ loss will be undefined.

3. An analyst is choosing between two different estimators, $\hat{\theta}_1$ and $\hat{\theta}_2$, to estimate a critical parameter θ . They have determined the following statistical properties for the estimators:

- **Estimator 1** ($\hat{\theta}_1$): $E[\hat{\theta}_1] = \theta$ and $Var(\hat{\theta}_1) = 9$
- **Estimator 2** ($\hat{\theta}_2$): $E[\hat{\theta}_2] = \theta + 0.5$ and $Var(\hat{\theta}_2) = 2$

Explain, which estimator is superior?

The estimator E1 is unbiased while E2 is biased; so it appears that E1 is better. However, note that E2 has much lower variance than E1 and effectively E2 will always have a lower loss than E1. Hence, E2 is the superior estimator.

4. A city analyst is tasked with predicting the number of bicycles rented from a station each hour. They start with a standard linear regression model but notice a problem: for certain times of day with low traffic, the model predicts -3 bicycles will be rented, which is physically impossible. Why is standard linear regression failing here? How do we fix this? (Hint: Think probabilistically!)

Standard linear regression fails because it assumes the output can be any real number and that its errors follow a normal (Gaussian) distribution. Count data (like the number of bikes) cannot be negative and often follows a different distribution, like a Poisson distribution, hence we require a GLM.

Advanced (4 marks)

1. Derive MLE linear regression when the assumption is made that the response variable has a Poisson distribution.

https://en.wikipedia.org/wiki/Poisson_regression

2. Derivation of the Closed-Form Solution for Ridge Regression.

The objective function for Ridge Regression is the sum of the Residual Sum of Squares (RSS) and an L2 regularization penalty. In matrix notation, this is:

$$J(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

The squared L2 norm of a vector $\mathbf{v}$ is $\mathbf{v}^T \mathbf{v}$. Applying this to our objective function:

$$\begin{aligned} J(\mathbf{w}) &= (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \\ &= (\mathbf{w}^T \mathbf{X}^T - \mathbf{y}^T) (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \\ &= \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{w}^T \mathbf{X}^T \mathbf{y} - \mathbf{y}^T \mathbf{X} \mathbf{w} + \mathbf{y}^T \mathbf{y} + \lambda \mathbf{w}^T \mathbf{w} \end{aligned}$$

Since $\mathbf{y}^T \mathbf{X} \mathbf{w}$ is a scalar, it is equal to its own transpose, $(\mathbf{y}^T \mathbf{X} \mathbf{w})^T = \mathbf{w}^T \mathbf{X}^T \mathbf{y}$. We can combine the two cross-terms:

$$J(\mathbf{w}) = \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{y}^T \mathbf{y} + \lambda \mathbf{w}^T \mathbf{w}$$

We differentiate $J(\mathbf{w})$ term by term using the following standard matrix calculus identities:

- $\nabla_{\mathbf{w}}(\mathbf{w}^T \mathbf{A} \mathbf{w}) = 2\mathbf{A} \mathbf{w}$ (for symmetric $\mathbf{A}$)
- $\nabla_{\mathbf{w}}(\mathbf{b}^T \mathbf{w}) = \mathbf{b}$

Note that $\mathbf{X}^T \mathbf{X}$ is a symmetric matrix.

$$\begin{aligned}\nabla_{\mathbf{w}} J(\mathbf{w}) &= \nabla_{\mathbf{w}} (\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{y}^T \mathbf{y} + \lambda \mathbf{w}^T \mathbf{w}) \\ &= 2\mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{X}^T \mathbf{y} + \mathbf{0} + 2\lambda \mathbf{I} \mathbf{w}\end{aligned}$$

Note: $\mathbf{w}^T \mathbf{w} = \mathbf{w}^T \mathbf{I} \mathbf{w}$, where $\mathbf{I}$ is the identity matrix. To find the minimum of the convex function $J(\mathbf{w})$, we set its gradient to the zero vector and solve for $\mathbf{w}$:

$$\begin{aligned}2\mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{X}^T \mathbf{y} + 2\lambda \mathbf{I} \mathbf{w} &= \mathbf{0} \\ \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y} + \lambda \mathbf{I} \mathbf{w} &= \mathbf{0}\end{aligned}$$

We group the terms containing $\mathbf{w}$:

$$\begin{aligned}\mathbf{X}^T \mathbf{X} \mathbf{w} + \lambda \mathbf{I} \mathbf{w} &= \mathbf{X}^T \mathbf{y} \\ (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}) \mathbf{w} &= \mathbf{X}^T \mathbf{y} \\ \mathbf{w}_{\text{ridge}} &= (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}\end{aligned}$$

This is the closed-form solution for the Ridge Regression coefficients. The term $\lambda \mathbf{I}$ ensures that the matrix is always invertible.

Bonus (2 marks)

1. A consistent estimator has the property that as the number of data points (which are used to estimate the parameters) increase, i.e. $N \rightarrow \infty$, the estimates converge in probability to the true parameters. Show that an unbiased estimator is not always consistent. Further, show that the OLS estimator is always consistent. You may use an example/counterexample to show the result.
2. +1 for scoring ≥ 16 marks in the exam.

We know an estimator is Unbiased when: $\mathbb{E}(\hat{\theta}_n) = \theta$, where $\hat{\theta}_n$ is the estimator for parameter θ . An estimator is said to be Consistent when:

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty$$

i.e., the estimator converges to the true parameter as the number of observations increases. Example for i.i.d. data $X_1, X_2, \dots, X_n$ each having mean to be θ and variance to be σ . Let's say our estimator be a sample from above data points i.e. X_1 :

$$\hat{\theta}_n = X_1$$

which is unbiased, since: $\mathbb{E}[\hat{\theta}_n] = \mathbb{E}[X_1] = \theta$. Now we will check consistency. As $n \rightarrow \infty$, our variance of $\hat{\theta}_n$ does not go to zero since there is no effect of N on the variance for this estimator the variance remains same. The OLS (Ordinary Least Squares) estimator is a consistent estimator. We know that the OLS estimator for β is: $\hat{\beta} = (X^T X)^{-1} X^T Y$ where $Y = X\beta + \epsilon$ (error term), and $\mathbb{E}[\epsilon] = 0$. To prove that OLS is unbiased we will find the expectation of $\hat{\beta}$

$$\begin{aligned}\mathbb{E}[\hat{\beta}] &= \mathbb{E}[(X^T X)^{-1} X^T Y] \\ &= (X^T X)^{-1} X^T \mathbb{E}[Y] \\ &= (X^T X)^{-1} X^T \mathbb{E}[X\beta + \epsilon] \\ &= (X^T X)^{-1} X^T X \mathbb{E}[\beta] + (X^T X)^{-1} X^T \mathbb{E}[\epsilon] \\ &= \beta + \mathbf{0} \\ \mathbb{E}[\hat{\beta}] &= \beta\end{aligned}$$

Thus, OLS is an unbiased estimator. Now to find if OLS is consistent, we will find the variance of the Sample Mean For data points $X_1, X_2, X_3, \dots, X_m$. having variance of each sample to be σ^2 . Variance of the average $\bar{X}_n$ is:

$$\begin{aligned}\text{Var}(\bar{X}_n) &= \frac{1}{N^2} \text{Var}(X_1 + X_2 + \dots + X_n) \\ &= \frac{1}{N^2} N \sigma^2 = \frac{\sigma^2}{N}\end{aligned}$$

As $N \rightarrow \infty$, $\text{Var}(\bar{X}_n) \rightarrow 0$, meaning $\hat{\beta}_n$ becomes more concentrated around β , showing consistency.